

Investigation of the Antibacterial Effects of Trimethoprim and Sulfamethoxazole Against *E. coli*

A disk-diffusion investigation of sequential folate-pathway inhibition and antibacterial synergy between two first-line antimicrobial agents.

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Undergraduate Research Report | Biomedical Sciences

Submitted as academic coursework; prepared here for portfolio presentation.

Abstract

AIMS

To investigate the effects of trimethoprim (TMP), sulfamethoxazole (SMX) and combined TMP+SMX treatment on the growth inhibition of *Escherichia coli* (*E. coli*) at increasing antibiotic concentrations.

METHODS AND RESULTS

Antibacterial activity was assessed using a disk diffusion assay. TMP and SMX were tested individually at 10 µg/mL, 100 µg/mL and 500 µg/mL, while combined TMP+SMX treatments contained equal concentrations of both antibiotics. Sterile water acted as a negative control and kanamycin acted as a positive control. Water controls produced no inhibition, whilst kanamycin consistently produced large inhibition zones. TMP generated larger inhibition zones than SMX at equivalent concentrations, while combined TMP+SMX treatments showed greater inhibition than individual treatments. Inhibition was quantified by measuring zone diameters (mm), and data were analysed using linear modelling, one-way ANOVA and post-hoc pairwise comparisons. Antibiotic treatment significantly affected inhibition zone diameter ($p < 0.001$), with the model explaining approximately 99% of the variation in antibacterial activity ($R^2 = 0.9927$).

CONCLUSIONS

Combined TMP+SMX treatments produced enhanced antibacterial activity against *E. coli*, supporting a synergistic interaction between the two antibiotics. Individually, TMP was more effective than SMX, and increasing antibiotic concentration increased bacterial growth inhibition. These findings highlight the clinical importance of combination therapies in improving antibacterial effectiveness and helping to reduce antibiotic resistance.

Keywords: antibiotic resistance; *Escherichia coli*; trimethoprim; sulfamethoxazole; synergistic antibiotic therapy; disk diffusion assay

Introduction

Antibiotic resistance is a major global health concern that reduces the effectiveness of antimicrobial treatments and increases the difficulty of controlling bacterial infections (WHO, 2014). The widespread use and misuse of antibiotics have accelerated the emergence of resistant bacterial strains, limiting the success of many commonly used antimicrobial drugs (Laxminarayan et al., 2013). As a result, there is increasing interest in identifying treatment strategies that improve antibacterial effectiveness and reduce the likelihood of resistance development. One approach commonly used in clinical treatment is combination antibiotic therapy, where two antibiotics are administered together to enhance antibacterial activity through synergistic interactions (Fischbach, 2011).

Escherichia coli (*E. coli*) is a gram-negative opportunistic pathogen associated with infections including urinary tract infections and septicaemia (Kaper, Nataro and Mobley, 2004). Increasing antibiotic resistance in *E. coli* presents a significant challenge for infection treatment and control (Tenailon et al., 2010).

Trimethoprim (TMP) and sulfamethoxazole (SMX) inhibit bacterial folic acid synthesis through different mechanisms. SMX inhibits dihydropteroic acid synthesis, while TMP inhibits dihydrofolate reductase, preventing bacterial DNA synthesis (Masters et al., 2003). Because these antibiotics target sequential stages within the same metabolic pathway, they are commonly administered together as co-trimoxazole. Combining antibiotics that target related pathways can produce synergistic antibacterial effects and may reduce the development of antibiotic resistance (Bushby, 1973; Fischbach, 2011).

The antibacterial activity of antibiotics can be assessed using a disk diffusion assay (Bauer et al., 1966; Clinical and Laboratory Standards Institute, 2020). Antibiotic-containing disks are placed onto bacteria-inoculated agar plates, and following incubation, zones of inhibition form where bacterial growth has been prevented. Larger inhibition zones indicate greater antibacterial effectiveness.

The aim of this study was to investigate the effects of TMP, SMX and combined TMP+SMX treatments on the inhibition of *E. coli* growth using a disk diffusion assay. It was hypothesised that combined TMP+SMX treatments would produce significantly larger zones of inhibition than the individual antibiotic treatments due to synergistic antibacterial activity between the two antibiotics.

Methods

Bacterial growth inhibition assay

The antibacterial activity of trimethoprim (TMP), sulfamethoxazole (SMX) and combined TMP+SMX treatments against *E. coli* was assessed using a disk diffusion assay on nutrient agar plates. TMP and SMX were tested individually at concentrations of 10 µg/mL, 100 µg/mL and 500 µg/mL. Combined treatments containing equal concentrations of both antibiotics (10+10 µg/mL, 100+100 µg/mL and 500+500 µg/mL) were also tested to investigate potential synergistic antibacterial interactions. Sterile water was used as a negative control and kanamycin (1000 µg/mL) was used as a positive control. Four independent experimental repeats were performed for each treatment group.

Stock antibiotic solutions of TMP and SMX were prepared and diluted with sterile distilled water to produce working concentrations of 10 µg/mL, 100 µg/mL and 500 µg/mL. Combination treatments were prepared by mixing equal volumes of TMP and SMX solutions at corresponding concentrations to obtain final combined treatments of 10+10 µg/mL, 100+100 µg/mL and 500+500 µg/mL. Sterile paper disks

were impregnated with the prepared antibiotic solutions prior to placement onto the inoculated agar surface.

E. coli was spread across nutrient agar plates to produce a bacterial lawn before antibiotic disks were applied. Plates were incubated at 37 °C for 24 h.

Measurements

Following incubation, antibacterial activity was quantified by measuring inhibition zone diameter (mm). Larger inhibition zones indicated greater antibacterial effectiveness. Measurements were recorded across four experimental repeats.

Summary statistics including mean, standard deviation (SD), sample size (n) and standard error (SE) were calculated for each treatment group. Data were visualised using boxplots and plots of treatment means with associated standard error bars, to compare antibacterial activity between treatments and concentrations.

Statistical analysis

All statistical analyses were conducted in R version 4.4.1 (R Core Team, 2024) using the tidyverse, readxl and emmeans packages. A linear model and one-way ANOVA were used to assess treatment effects on inhibition zone diameter, followed by post-hoc pairwise comparisons using estimated marginal means. Statistical significance was assessed at $\alpha = 0.05$.

Results

Analysis of the model data showed that antibiotic treatment significantly affected inhibition of *E. coli* growth (ANOVA; $F = 446.66$, d.f. = 10,33; $p < 0.001$). The fitted linear model accounted for approximately 99% of the variation in inhibition zone diameter between treatment groups ($R^2 = 0.9927$), indicating that antibacterial activity was strongly associated with antibiotic treatment.

As shown in Figure 1, inhibition zone diameter generally increased with increasing antibiotic concentration for all treatments (Bushby and Hitchings, 1968). TMP+SMX combination treatments generated the largest zones of inhibition; however, the magnitude of this increase became proportionally smaller at higher antibiotic concentrations, where inhibition zones for both treatments approached similarly large diameters. TMP treatments resulted in greater inhibition than SMX treatments at all concentrations tested.

Sterile water produced no inhibition, while kanamycin generated large inhibition zones, confirming assay validity (Figure 1). Post-hoc pairwise comparisons identified several significant differences between treatment groups (Figure 1). TMP treatment produced significantly larger inhibition zones than equivalent SMX treatment concentrations at 10 µg/mL ($p = 0.0092$), 100 µg/mL ($p < 0.0001$) and 500 µg/mL ($p < 0.0001$). In addition, combined TMP+SMX treatments produced significantly greater inhibition than TMP-only treatments at concentrations of 100+100 µg/mL ($p < 0.0001$) and 500+500 µg/mL ($p < 0.0001$).

Overall, combined TMP+SMX treatments exhibited greater antibacterial activity than several individual antibiotic treatments, with the strongest inhibition observed at the highest combined concentration tested.

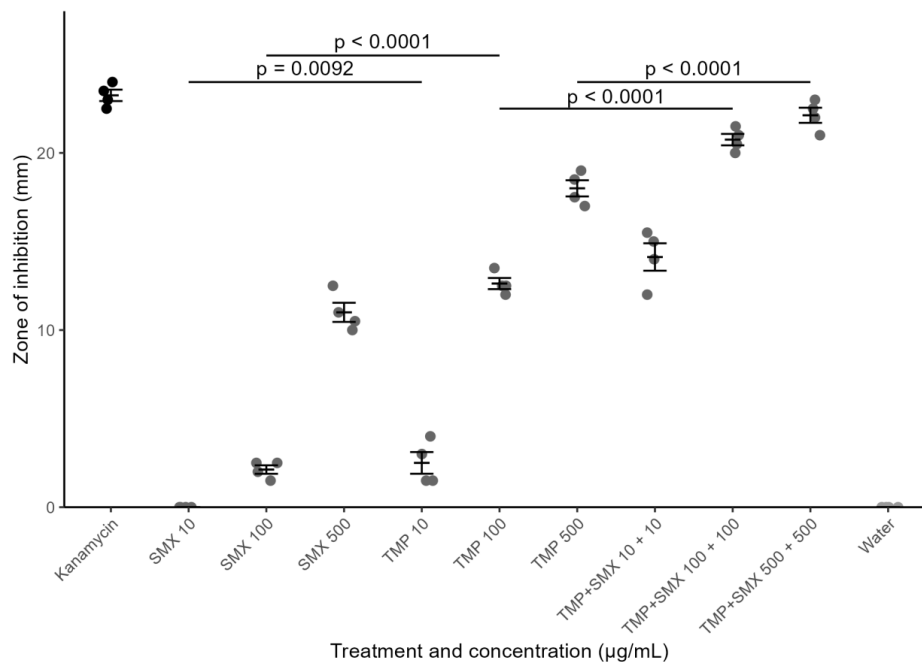


Figure 1. Growth inhibition of *E. coli* following treatment with individual and combined antibiotics. Points represent individual replicates ($n = 4$) and horizontal bars indicate mean $\pm$ standard error (SE). TMP (trimethoprim) and SMX (sulfamethoxazole) were tested individually at 10 $\mu\text{g/mL}$, 100 $\mu\text{g/mL}$ and 500 $\mu\text{g/mL}$, and in combination at equal concentrations (10+10 $\mu\text{g/mL}$, 100+100 $\mu\text{g/mL}$ and 500+500 $\mu\text{g/mL}$). Sterile water and kanamycin were used as negative and positive controls, respectively. Horizontal comparison bars indicate significant pairwise differences identified by post-hoc estimated marginal means analysis.

Discussion

Summary of findings

This study investigated the antibacterial effects of TMP, SMX and combined TMP+SMX treatments against *E. coli* using a disk diffusion assay. Antibiotic treatment significantly affected inhibition zone diameter, with combined TMP+SMX treatments generally producing larger zones of inhibition than individual antibiotic treatments alone. These findings support the hypothesis that combined TMP+SMX treatment would produce greater antibacterial activity than either antibiotic used independently. Overall, the investigation successfully addressed the study aim by demonstrating that combined TMP+SMX treatments produced greater antibacterial inhibition of *E. coli* than several individual antibiotic treatments.

Antibacterial effectiveness increased with increasing antibiotic concentration across both individual and combined treatments, indicating a concentration-dependent response. TMP consistently produced larger inhibition zones than equivalent SMX concentrations, suggesting greater antibacterial effectiveness against *E. coli* under the experimental conditions used. In contrast, the TMP+SMX combination treatments produced the greatest overall inhibition, particularly at higher concentrations. However, although synergistic effects were observed across all combined treatments, the difference in inhibition between lower- and higher-concentration TMP+SMX treatments was relatively small compared with the increase in antibiotic concentration. This suggests that increasing concentration beyond a certain level may provide diminishing increases in antibacterial activity. These findings contribute to understanding how combination antibiotic therapy may improve antibacterial effectiveness while potentially reducing reliance on high individual antibiotic concentrations, which may

help to limit selective pressure for antibiotic resistance development. These findings are consistent with previous studies demonstrating enhanced antibacterial activity when TMP and SMX are administered together as co-trimoxazole (Bushby and Hitchings, 1968; Bushby, 1969). However, the comparatively small increase in inhibition observed at higher combination concentrations suggests that synergistic effects may become less pronounced once near-maximal bacterial inhibition has been achieved.

Mechanistic interpretation

The enhanced inhibition observed in the combined treatments is likely explained by the complementary mechanisms of action of TMP and SMX. Both antibiotics interfere with bacterial folic acid synthesis but target different stages within the metabolic pathway. Sulfamethoxazole inhibits the synthesis of dihydropteroic acid, whereas trimethoprim inhibits dihydrofolate reductase, preventing the formation of tetrahydrofolate required for DNA synthesis. Sequential inhibition of the same biosynthetic pathway is known to enhance antibacterial activity and can produce synergistic effects (Masters et al., 2003). The larger inhibition zones observed for the TMP+SMX treatments are therefore consistent with the established synergistic interaction between these antibiotics. However, although the observed increases in inhibition zone diameter are consistent with synergistic antibacterial interactions, definitive confirmation of synergy would require quantitative approaches such as checkerboard or fractional inhibitory concentration (FIC) analyses.

Although TMP+SMX treatments produced greater inhibition than TMP alone, the relative increase became less pronounced at higher antibiotic concentrations, suggesting antibacterial activity approached a maximal inhibitory effect. Similar concentration-dependent reductions in synergistic effects have been reported in antimicrobial combination therapies, where bacterial inhibition increasingly approaches maximal levels under high individual drug concentrations (Torella, Chait and Kishony, 2010).

Assay validity

The control treatments confirmed that the assay functioned correctly. The sterile water control produced no detectable inhibition of bacterial growth, demonstrating that inhibition zones resulted from antibiotic activity rather than the paper disks or solvent. In contrast, kanamycin produced consistently large inhibition zones, confirming that the *E. coli* strain remained susceptible to antibacterial treatment and that the experimental conditions supported normal bacterial growth and antibiotic diffusion.

Limitations

Several limitations should be considered. Only a single bacterial species was tested, limiting generalisation to other pathogens. Additionally, although four biological replicates were included, statistical power remained limited. Disk diffusion assays provide an indirect measure of antibacterial activity because inhibition zone diameter is influenced by antibiotic diffusion through agar (Balouiri, Sadiki and Ibnsouda, 2016). The study also did not distinguish between bacteriostatic and bactericidal effects. Furthermore, because the investigation was conducted entirely *in vitro*, the findings may not fully reflect antibacterial effectiveness *in vivo*, where host immune responses, tissue penetration and antibiotic pharmacokinetics influence treatment outcomes. More quantitative approaches, such as MIC, checkerboard or time-kill assays, would allow more precise evaluation of synergistic interactions.

Future directions

Future work could investigate the effects of TMP and SMX combinations against a wider range of Gram-negative and Gram-positive bacterial species, including antibiotic-resistant strains. Additional studies using MIC-based methods and time-kill assays would also provide greater insight into the extent and mechanism of synergistic antibacterial activity. Furthermore, investigating different TMP:SMX concentration ratios may help determine the optimal combination required to maximise antibacterial effectiveness while minimising antibiotic usage.

Conclusion

In conclusion, combined TMP+SMX treatments produced greater inhibition of *E. coli* growth than several individual antibiotic treatments, while increasing antibiotic concentration generally enhanced antibacterial effectiveness. However, the relatively small difference in synergistic inhibition between lower- and higher-concentration combination treatments suggests that maximal antibacterial activity may be approached at comparatively moderate concentrations. Overall, the findings support the established synergistic interaction between TMP and SMX and highlight the continued importance of combination antibiotic therapies in improving antibacterial activity. These results further demonstrate the potential value of combination antibiotic therapies in addressing the growing challenge of antimicrobial resistance.

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